Chetan Goenka

Berkeley, CA | (512) 363-2290 | cgoenka@berkeley.edu | cgoenka.me | github.com/c-goenka | linkedin.com/in/cgoenka

EDUCATION

University of California, Berkeley — Berkeley, CA

Master of Science in Electrical Engineering & Computer Science — Expected May 2026

Bachelor of Arts in Computer Science, GPA: 3.90/4.0, Graduated with Honors — Dec 2024

Relevant Coursework: Efficient Algorithms, Data Structures, Artificial Intelligence, Deep Neural Networks, User Interface Development, Internet Architecture and Protocol, Computer Architecture, Machine Learning & Human Behavior

SKILLS

Programming Languages: Python, Java, SQL, TypeScript/JavaScript, C/C++, C#, HTML, CSS, Swift, R, Bash, Assembly
Frameworks/Tools: Pandas, NumPy, Git, React, Next.js, Tailwind, OpenAI/Claude, AWS, PostgreSQL, Figma, Pytest, Docker

RELEVANT EXPERIENCE AND PROJECTS

UC BERKELEY EECS DEPARTMENT - HCI Research Lab — Berkeley, CA
Engineering Design Scholar & Research Assistant — June 2024 – Present

- Engineered a web-based coding environment using React, CodeMirror, and OpenAI API, delivering dynamic, context-aware code explanations (line-by-line, block-wise, and data-flow) to enhance developer comprehension of AI-generated code
- Designed and implemented structured JSON metadata extraction with Zod, ensuring accurate line mappings, logical groupings, and data-flow tracking, enabling dynamically linked LLM-generated explanations and interactive overlays
- Extended CodeMirror's annotation capabilities by building custom DOM-based overlays for code elements, hover-triggered tooltips, and click-driven interactions, improving code navigation and readability through dynamic in-editor explanations
- Conducted pilot study with developers to evaluate tool effectiveness, analyzing user interaction data and feedback to validate design decisions and measure comprehension improvements

PYTHON COMPILER — Jan 2024 – May 2024

- Developed a compiler using Java for ChocoPy, a statically typed dialect of Python, implementing key stages including lexical analysis, syntax analysis, and optimized code generation for RISC-V assembly
- Constructed a code generator to translate type-annotated ASTs into efficient RISC-V assembly code, integrating techniques such as constant propagation, dead code elimination, and register allocation, reducing runtime by 40%
- Implemented list operations, function calls, and control flow structures, ensuring correct execution of ChocoPy programs on a RISC-V simulator, and constructed an integrated semantic analyzer and type checker to maintain well-typed programs

UC BERKELEY EECS DEPARTMENT - Programming Course (Data 100) — Berkeley, CA
Undergraduate Course Staff — Aug 2023 – May 2024

- Provided weekly programming guidance to 25+ students in Python and data manipulation libraries, helping debug complex programming projects and resolve syntax errors, logic bugs, and algorithmic challenges
- Developed comprehensive troubleshooting strategies and technical study materials for common programming challenges, facilitating collaborative problem-solving in office hours, online forums, and discussion sections
- Evaluated 5000+ student assignments using structured rubrics and provided detailed code feedback, ensuring adherence to coding standards and technical requirements through systematic review processes across multiple semesters

ORACLE CORPORATION — Austin, TX
Data Science Intern — Jun 2022 – Aug 2022

- Developed a customer scoring system using Python, pandas, and scikit-learn with regression and AdaBoost algorithms, collaborating with team members to increase marketing efficiency by 25% across enterprise accounts
- Enhanced data preprocessing and model deployment pipeline for production systems through data cleaning and feature engineering techniques, collaborating with senior engineers to decrease overall run time by 18%
- Analyzed marketing campaign data from hundreds of customer interactions using Python and SQL to extract key insights, creating visualizations and reports to communicate findings to sales and marketing teams